SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

PUZZLE BOOK

Wrangle It, Clean It, Chart It: 40 Puzzles Across the Whole Syllabus

UDSC-402: Data Analysis and Visualization	Instructor: Dr. D Bhanu Prakash	Date: 24-10-2026

Puzzle 1 · Word Hunt Q1–10. Answer the question, then find its key word in a 15 × 15 grid.

Puzzle 2 · Crossword Q11–20. Clues are the original questions.

Puzzle 3 · Fact or Fib Q21–30. Correct choices spell a hidden 10-letter word.

Puzzle 4 · Match-Up Grid Q31–40. Tick-grid logic puzzles.

Name: _____ Date: _____ Score: _____ / 40

Puzzle 1 — Word Hunt

Answer each multiple-choice question. The key word of every correct answer is hidden in the grid — across, down or diagonally (↘ ↗). Circle it and note the question number.

D	W	K	H	C	Y	V	L	W	Q	S	I	P	G	F
I	M	P	U	T	I	N	G	D	J	D	J	C	K	I
T	I	X	P	C	P	Y	P	N	D	K	Q	M	A	G
E	F	I	Q	O	L	F	I	E	E	F	D	J	N	I
R	V	M	K	B	S	I	N	R	Y	R	C	I	L	J
A	D	C	W	X	A	I	U	F	A	Q	H	T	T	K
T	T	P	V	B	F	S	T	O	O	C	K	B	P	P
I	B	P	Q	E	O	L	B	I	I	Q	B	S	J	U
V	L	A	R	L	F	H	X	R	O	I	H	Z	P	I
E	U	S	C	U	S	Q	N	F	B	N	Q	J	X	T
P	Y	N	S	A	G	E	E	I	G	H	T	Y	F	A
F	E	V	D	Z	P	L	A	Y	F	A	I	R	H	Z
E	D	M	S	S	D	L	E	O	Y	Z	U	J	B	J
E	A	Z	P	V	I	W	R	A	P	P	E	R	U	Y
H	K	K	N	H	F	Z	C	L	U	T	Q	A	C	L

Q1. Which of these is NOT one of the four steps of the applied machine learning process? (a) Define Problem (b) Prepare Data (c) Publish ____ (d) Evaluate Models *Hint: Find the one word missing in (c).*

Q2. Recursive Feature Elimination (RFE) is an example of which category of feature selection? (a) Filter (b) Wrapper (c) Intrinsic (d) Unsupervised

Q3. Which imputation technique is also known as MICE — Multivariate Imputation by Chained Equations? (a) Statistical (b) KNN (c) Iterative (d) Listwise deletion

Q4. According to Principles of Data Wrangling, what share of an analyst's time goes on wrangling? (a) 10–20% (b) 30–40% (c) 50–80% (d) 90–95% *Hint: Hunt the spelled-out upper limit of the correct range.*

Q5. Most data projects come to an end in which stage? (a) Raw (b) Refined (c) Production (d) Archival

Q6. Which transformation type results in a net addition of information to a dataset? (a) Structuring (b) Enriching (c) Cleaning (d) Profiling *Hint: Hunt the -ING form.*

Q7. Which is a cleaning transformation (not structuring/enriching)? (a) Pivoting (b) Joining on a key (c) ____ missing values (d) Extracting a substring *Hint: Find the verb (-ING form) that fills the blank in (c).*

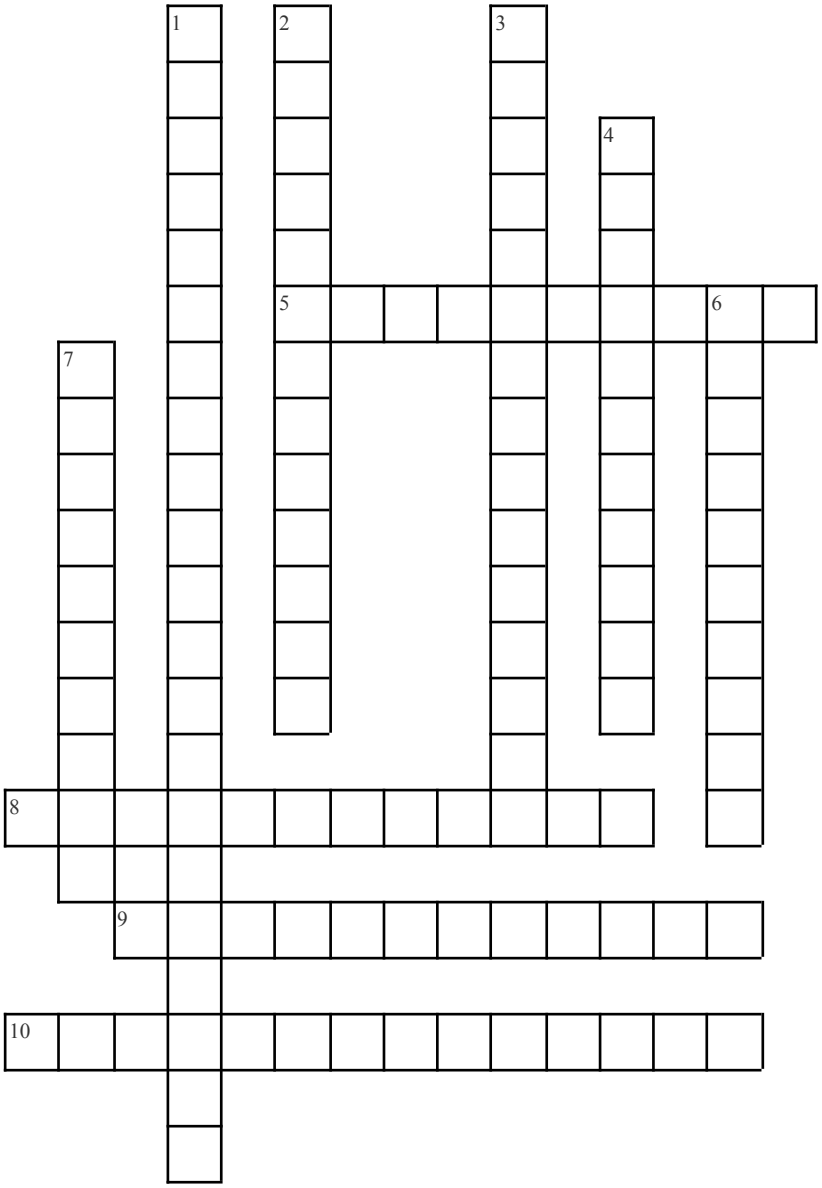
Q8. Per Stephen Few, which two visual attributes encode quantitative values accurately? (a) Hue & saturation (b) Area & colour intensity (c) Length & 2-D ____ (d) Shape & orientation *Hint: Find the word that completes (c).*

Q9. Who invented the bar graph and — on an off day — also the pie chart? (a) Descartes (b) ____ (c) Tufte (d) Cleveland *Hint: Find the surname of the right person.*

Q10. The relative perceptual strength of the Gestalt grouping principles is best stated as: (b) ____ > connection > proximity and similarity *Hint: Find the strongest principle.*

Puzzle 2 — Crossword

Each clue is a fill-in-the-blank question. Write the answer in the grid, with no spaces or hyphens between words. The number in brackets is the letter count.



ACROSS →

5. _____ is a characteristic of an object that reveals how it is supposed to be used and makes it easy to use that way. **(10)**

8. A column in which every row holds exactly the same value is called a _____ predictor. **(12)**

9. Tufte's measure of the ink that presents information compared with the total ink is the _____. **(12)**

10. A concise KPI target that both predicts the desired impact and is actionable is called a _____. **(14)**

DOWN ↓

1. As input features grow, data becomes an increasingly sparse, unrepresentative sample of the feature space — this is known as the _____. **(21)**

2. ____ means data must adhere to structural and syntactic constraints in order to be ingested at all. **(13)**

3. The transformation of raw data into a form more suitable for modelling is called _____. **(15)**

4. _____ is the problem in which information about the holdout dataset is made available to the model during training. **(11)**

6. A map produced by applying colour intensity directly to geographical regions is called a __ map. **(10)**

7. A sample that deliberately covers every group or "stratum" in the data is called a _____ sample. **(10)**

Puzzle 3 — Fact or Fib? Secret Word

Decide whether each statement is TRUE or FALSE and circle the letter beside your choice. Copy the circled letters, in order, into the boxes at the bottom to reveal a 10-letter secret word (a clue to what this whole subject is for!).

No.	Statement	TRUE →	FALSE →
21	Data cleaning is typically performed after feature engineering and dimensionality reduction.	**Z**	**V**
22	In k-fold cross-validation, the data preparation transform must be fit inside each fold using only that fold's training portion.	**I**	**V**
23	Variables produced by dimensionality reduction remain directly interpretable in terms of the original input variables.	**R**	**S**
24	ETL is one type of data wrangling.	**U**	**O**
25	Unions stack datasets vertically, whereas joins link them horizontally.	**A**	**Q**
26	Indirect, human-mediated value is a prerequisite for direct, automated value.	**L**	**U**
27	Profiling is performed only at the very end of a wrangling project, after all transformations are complete.	**P**	**I**
28	A single set of values occupying one column or one row qualifies as a table.	**M**	**Z**
29	When bars encode the values, the quantitative scale may safely begin above zero.	**Q**	**E**
30	In a time-series graph, time must run horizontally from left to right along the X axis.	**S**	**O**

Secret word:

(Letters from Q21 … Q30 go into boxes 1 … 10.)

Puzzle 4 — Match-Up Grid

Read the descriptions in Column II (a–e). Put a tick (✔) in the one box in each row that correctly matches the Column I item. Every row and every column gets exactly one tick — use that to your advantage!

Set A — Questions 31–35: Match the term with its description

(a)	Model-based automatic outlier detection
(b)	Intrinsic (supervised) feature selection
(c)	Visual data preparation tool launched in 2012
(d)	Statistical imputation using mean / median / mode
(e)	Structure is not enforced until the data is used

Column I ↓ / Column II →	(a)	(b)	(c)	(d)	(e)
31. LASSO regression / decision trees					
32. Local Outlier Factor					
33. SimpleImputer					
34. Schema-on-read					
35. Trifacta Wrangler					

Set B — Questions 36–40: Match the display with the situation it suits

(a)	Two sorted bar graphs side by side — an intuitive alternative to a scatter plot
(b)	Correlation between two quantitative variables
(c)	Distribution — median, midspread, quartiles and range
(d)	Ranking, or a time series with irregular intervals where the scale cannot start at zero
(e)	A series of related graphs used instead of adding a third dimension

Column I ↓ / Column II →	(a)	(b)	(c)	(d)	(e)
36. Scatter plot					
37. Box plot					
38. Dot plot					
39. Table lens					
40. Small multiples					

Answer Key

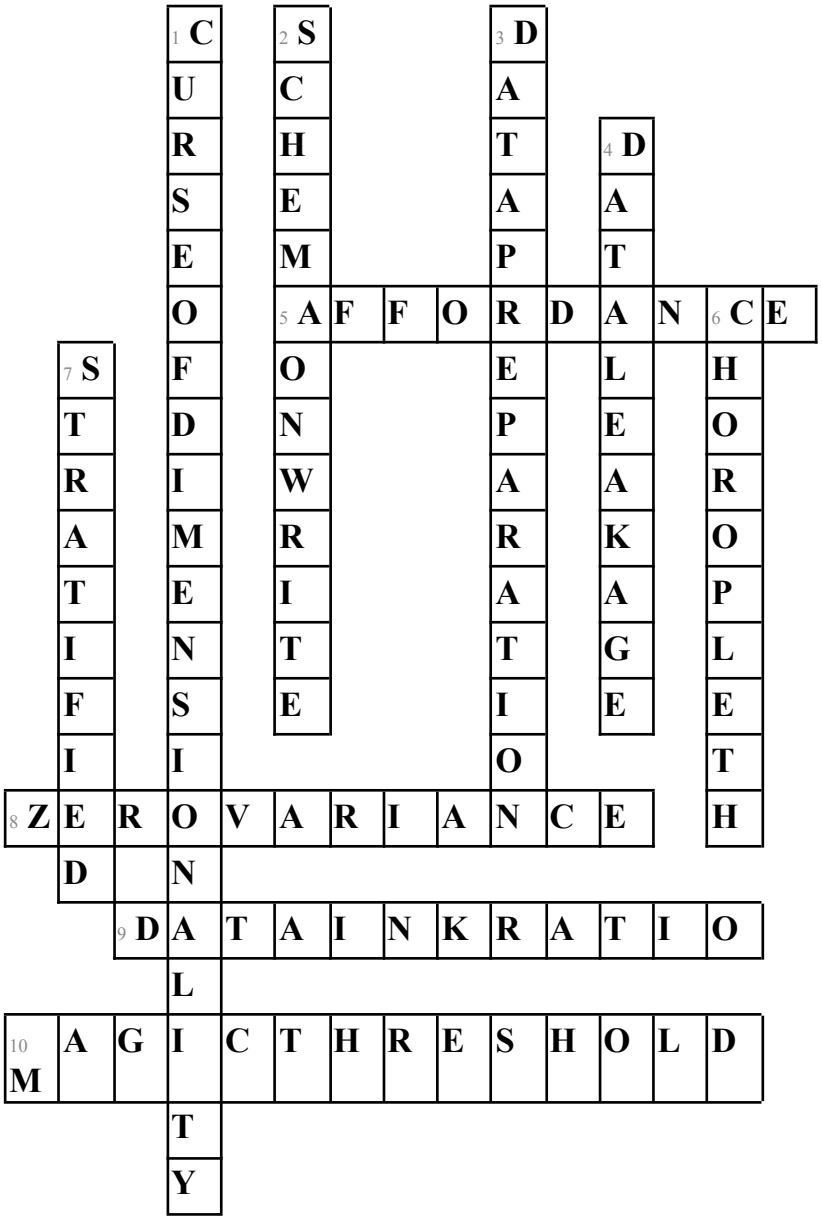
For the teacher / self-check.

Puzzle 1 — Word Hunt (highlighted words)

D	W	K	H	C	Y	V	L	W	Q	S	I	P	G	F
I	M	P	U	T	I	N	G	D	J	D	J	C	K	I
T	I	X	P	C	P	Y	P	N	D	K	Q	M	A	G
E	F	I	Q	O	L	F	I	E	E	F	D	J	N	I
R	V	M	K	B	S	I	N	R	Y	R	C	I	L	J
A	D	C	W	X	A	I	U	F	A	Q	H	T	T	K
T	T	P	V	B	F	S	T	O	O	C	K	B	P	P
I	B	P	Q	E	O	L	B	I	I	Q	B	S	J	U
V	L	A	R	L	F	H	X	R	O	I	H	Z	P	I
E	U	S	C	U	S	Q	N	F	B	N	Q	J	X	T
P	Y	N	S	A	G	E	E	I	G	H	T	Y	F	A
F	E	V	D	Z	P	L	A	Y	F	A	I	R	H	Z
E	D	M	S	S	D	L	E	O	Y	Z	U	J	B	J
E	A	Z	P	V	I	W	R	A	P	P	E	R	U	Y
H	K	K	N	H	F	Z	C	L	U	T	Q	A	C	L

Q	Correct answer	Option	Word in grid
1	Publish Dashboard	(c)	DASHBOARD
2	Wrapper	(b)	WRAPPER
3	Iterative imputation	(c)	ITERATIVE
4	50–80%	(c)	EIGHTY
5	Refined stage	(b)	REFINED
6	Enriching	(b)	ENRICHING
7	Imputing missing values	(c)	IMPUTING
8	Length and 2-D position	(c)	POSITION
9	William Playfair	(b)	PLAYFAIR
10	enclosure > connection > proximity and similarity	(b)	ENCLOSURE

Puzzle 2 — Crossword solution



5A (Q18): AFFORDANCE · 8A (Q12): ZERO-VARIANCE · 9A (Q19): DATA-INK RATIO · 10A (Q17): MAGIC THRESHOLD · 1D (Q14): CURSE OF DIMENSIONALITY · 2D (Q15): SCHEMA-ON-WRITE · 3D (Q11): DATA PREPARATION · 4D (Q13): DATA LEAKAGE · 6D (Q20): CHOROPLETH · 7D (Q16): STRATIFIED

Puzzle 3 — Fact or Fib

Q21 FALSE (V) · Q22 TRUE (I) · Q23 FALSE (S) · Q24 TRUE (U) · Q25 TRUE (A) · Q26 TRUE (L) · Q27 FALSE (I) · Q28 FALSE (Z) · Q29 FALSE (E) · Q30 TRUE (S)

Secret word: VISUALIZES

Puzzle 4 — Match-Up Grid

Q31–35: 31 – (b) · 32 – (a) · 33 – (d) · 34 – (e) · 35 – (c)

Q36–40: 36 – (b) · 37 – (c) · 38 – (d) · 39 – (a) · 40 – (e)

Puzzle 4 — filled grids

Set A — Questions 31–35: Match the term with its description

(a)	Model-based automatic outlier detection
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(c)	Visual data preparation tool launched in 2012
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(e)	Structure is not enforced until the data is used

Column I ↓ / Column II →	(a)	(b)	(c)	(d)	(e)
31. LASSO regression / decision trees		✔			
32. Local Outlier Factor	✔				
33. SimpleImputer				✔	
34. Schema-on-read					✔
35. Trifacta Wrangler			✔		

Set B — Questions 36–40: Match the display with the situation it suits

(a)	Two sorted bar graphs side by side — an intuitive alternative to a scatter plot
(b)	Correlation between two quantitative variables
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Column I ↓ / Column II →	(a)	(b)	(c)	(d)	(e)
36. Scatter plot		✔			
37. Box plot			✔		
38. Dot plot				✔	
39. Table lens	✔				
40. Small multiples					✔